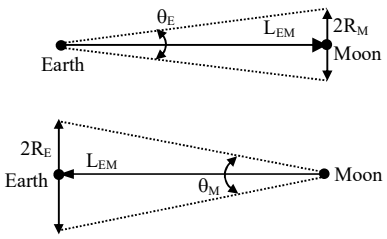
Exercise 4

A bi-directional communication link must be established between Earth and Moon. Each station employs a single antenna for transmitting and receiving. The link Earth→ Moon operates at f_{TE}=6 GHz and the opposite (Moon→Earth) at f_{TM}=4 GHz. The antennas must be dimensioned in order that their beam width correspond to the solid angle under which the moon is seen by the earth (earth antenna) and the earth is seen by the moon (moon antenna). The following picture defines the geometry and the parameters of the links.



L_{EM} (Earth-Moon distance): 384400 Km
R_E (Earth radius): 6371 Km
R_M (Moon radius): 1378 Km
θ_E Beam width of the earth antenna
θ_M Beam width of the moon antenna

a) Evaluate the beam width of the two antennas and the corresponding directivity gain (use the approximate formulas). Assuming that dish antennas are employed, evaluate their diameter. Dimension the antennas at the transmitting frequencies(f_{TE}, f_{TM}); assume the aperture efficiency e_a= 0.65 and the efficiency factor η=0.7 for both antennas.

b) Assuming that the power transmitted from the Earth is 500W and that transmitted from the Moon is 50W evaluate the power of the received signals at the two stations

c) The SNR (signal-to-noise ratio) at the input of the receivers must be at least 20 dB. The signal bandwidth is 100 MHz and the noise temperature of the antennas is 50 K (the transmitted powers are the same as in previous point). Compute the maximum values of the equivalent noise temperature at input of the receivers in order to meet the SNR requirement.

Use K=-228.6 dBW/(Hz· K) (Boltzmann Constant)

Solution:

The beam width angles are immediately derived from the figures:

$$\theta_E = 2\tan^{-1}\left(\frac{R_M}{L_{EM}}\right) = 0.411°, \qquad \theta_M = 2\tan^{-1}\left(\frac{R_E}{L_{EM}}\right) = 1.9°$$

The directivity gain of the antennas is obtained with the following formula: $\cos(\theta/2) = 1 - 2/D$:

$$D_{ET} = \frac{2}{1-\cos(\theta_E/2)} = 3.113\cdot 10^5 \,(54.93 \text{ dB}), \quad D_{MT} = \frac{2}{1-\cos(\theta_M/2)} = 1.4565\cdot 10^4 \,(41.63 \text{ dB})$$

The dishes radius is then obtained from:

$$\left(\frac{d}{\lambda}\right)^2 = \frac{D\cdot\eta}{\pi^2\cdot e_a}$$

For the earth antenna $\lambda_E = 3\cdot 10^8/6\cdot 10^9 = 0.05$m while for the moon antenna $\lambda_M = 3\cdot 10^8/4\cdot 10^9 = 0.075$m. Then:

$$d_E = \frac{\lambda_E}{\pi}\sqrt{\frac{\eta D_{ET}}{e_a}} = 9.21\text{m}, \quad d_M = \frac{\lambda_M}{\pi}\sqrt{\frac{\eta D_{MT}}{e_a}} = 2.99\text{m}$$

The gain of the antennas (transmission):

$G_{ET} = D_{ET} + 10\cdot\log(\eta) = 53.38$ dB, $G_{MT} = D_{MT} + 10\cdot\log(\eta) = 40.08$ dB

The gain of antenna in reception is computed from the dish formulas:

$$G_{ER} = e_a\left(\frac{\pi d_E}{\lambda_M}\right)^2 = 49.86 \text{ dB}, \quad G_{MR} = e_a\left(\frac{\pi d_M}{\lambda_E}\right)^2 = 43.6 \text{ dB}$$

Computation of the links attenuation:

Earth→Moon (6Hz): $A_{EM} = 20\cdot\log(4\pi L_{EM}/\lambda_E) = 219.7$ dB

Moon→ Earth (4Hz): $A_{ME} = 20\cdot\log(4\pi L_{EM}/\lambda_M) = 216.2$ dB

The received power at the two receiving stations is obtained with the Friis Equation:

$P_{RM} = P_{TE} - A_{EM} + G_{ET} + G_{MR} = -95.72$ dBW, $P_{RE} = P_{TM} - A_{EM} + G_{MT} + G_{ER} = -109.24$ dBW

The noise at the antennas output is determined by the SNR (20 dB) and the received signals power:

$N_E = P_{RE} - SNR = -129.24$ dBW, $N_M = P_{RM} - SNR = -115.72$ dBW.

The noise power can be expressed as $N = KT_{sys}B$, with $T_{sys} = T_a + T_{rec}$. It is then obtained:

$T_{eq,E} = 86.22$ K → $T_{rec,E} = 36.2$ K

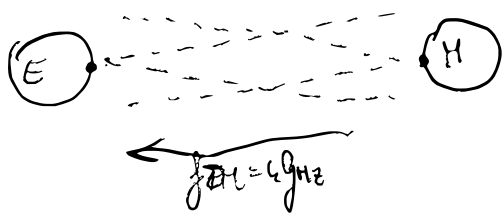
$T_{eq,M} = 1939.7$ K → $T_{rec,M} = 1889.7$ K

EXERCISE on ANTENNAS (9/2/2015) 16:33

EARTH ↔ MOON

BOTH ANTENNAS are AUTOMATICALLY ALIGNED

The two antennas MUST cover the other body's area within 3 dB



(FDD)

$L_{EARTH-MOON} = 384'400 Km$

$R_{EARTH} = 6371 Km$

$R_{MOON} = 1378 Km$

$dB_i = 10 \log 10$

$\alpha = \frac{HPBW}{2}$

① HPBW of EACH? DIRECTIVITY (APPROX)?

$\vartheta_{BW}_{EARTH} = \alpha \tan \frac{R_{MOON}}{L_{E-M}} = 0.2054°$ $HPBW_E = 0.41°$ $D_{E_{EARTH}} = \frac{2}{1-\cos\vartheta} = 311 \cdot 10^3 = 54.9 dBi$

$\cos\vartheta_{BW} \approx 1 - \frac{2}{D_x}$

$\vartheta_{BW}_{MOON} = \alpha \tan \frac{R_{EARTH}}{L_{E-M}} = 0.9495°$ $HPBW_H = 1.899°$ $D_{H_{MOON}} = 14.5 \cdot 10^3 = 41.6 dBi$

② DIAMETER of DISH ANTENNAS (AT THE TX FREQ; $e_A = 0.65$; $\eta = 0.7$)

$A_{PHYS} = \pi r^2$ $A_e = \frac{\lambda^2}{4\pi} G = \frac{\lambda^2}{4\pi} D \cdot \eta = e_A A_{PHYS} = e_A \cdot \pi \left[\frac{\phi}{2}\right]^2$

$G_E = 217.7 \cdot 10^3 = D_{TdBi} + 10 \log 0.7 = 53.4 dBi$

@6GHz: $A_e = 43.3 m^2 \rightarrow A_{PHYS} = 66.6 m^2 \rightarrow \phi_{EARTH} = 2 \cdot \sqrt{\frac{A_{PHYS}}{\pi}} = 9.2 m$

$G_H = 40 dBi = 10'000 \rightarrow \phi_{MOON} = \frac{\lambda}{\pi} \sqrt{\frac{D\eta}{e_A}} = 2.98 m \approx 3 m$ (@4GHz)

③ $P_{TX\,EARTH} = 500\,W$ $\quad P_{TX\,MOON} = 50\,W$

→ $P_{RX\,MOON}$, $P_{RX\,EARTH}$? (Assume perfect alignment)

$$P_{RX} = \underbrace{P_{TX_{dB}} + G_{TX_{dBi}}}_{EIRP} + 20\log\frac{\lambda}{4\pi R} + G_{RX\,dBi}$$

$$G_{MOON}^{6GHz} = \frac{A_e}{(\lambda^2/4\pi)} = \frac{\frac{\lambda_6^2}{\pi^2}\cdot G_{MOON}^{4GHz}}{\frac{\lambda_6^2}{4\pi}}$$

ISSUE:
$\cos\theta_{3dB}^{MOON} = 1 - \frac{2}{D_m} = 1 - \frac{2R}{G_{MOON}^{6}} \Rightarrow \theta_{3dB}^{6GHz} = 0.64^\circ$ → DOESN'T COVER WHOLE EARTH

BUT LET'S NEGLECT THE ISSUE

$$G_{MOON}^{6GHz} = 20\log\frac{6GHz}{4GHz} + G_{MOON}^{4G} = 3.5dB + 40dBi$$

$P_{RX_{MOON}} = 27dBW + 53.4dBi - 219.7dB + 43.5dBi = -95.8dBW$

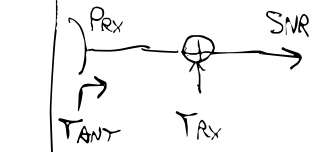
@6GHz $\quad = 263\,pW \quad = -65.8dBm$

$$G_{EARTH\,4GHz} = G_{EARTH\,6GHz} + 20\log\frac{4}{6}$$

$P_{RX\,EARTH} = 17dBW + 40dBi - 216.2dB + 49.9dBi = -109.3dBW$

@4GHz $\quad = 11.7\,pW \quad = -79.3dBm$

$SNR \geq 20dB$ $\quad BW = 100MHz$ $\quad T_{ANT}^{BOTH} = 50K \rightarrow T_{eq\,RX} = ?$ $\quad \left[\begin{array}{l} K = -228.6\,\frac{dBW}{Hz\,K} \\ K = 1.38\cdot10^{-23}\,\frac{W}{Hz\,K} \end{array}\right.$



$$SNR = \frac{P_{RX}}{K(T_{ANT}+T_{RX})BW}$$

$P_{NOISE@MOON} \leq -65.8dBm - 20dB = -85.8dBm = 2.63\,pW = KT_{ANT}BW + P_{N\,RX}$ (0.07 pW)

$\Rightarrow P_{N\,RX} \leq 2.56\,pW \rightarrow T_{RX} \leq 1855\,K$

[IN SOLUTIONS: 1889.7 K BUT THIS IS OK DUE TO ROUNDING]

$P_{NOISE@EARTH} \leq -99.3dBm \Rightarrow T_{SYS} = (T_0 + T_{RX}) = 85K \rightarrow T_{RX} = 35K$

30m